

# A Deep Learning Framework for Deriving Electrical Properties in Epithelial Membranes through Impedance Spectroscopy

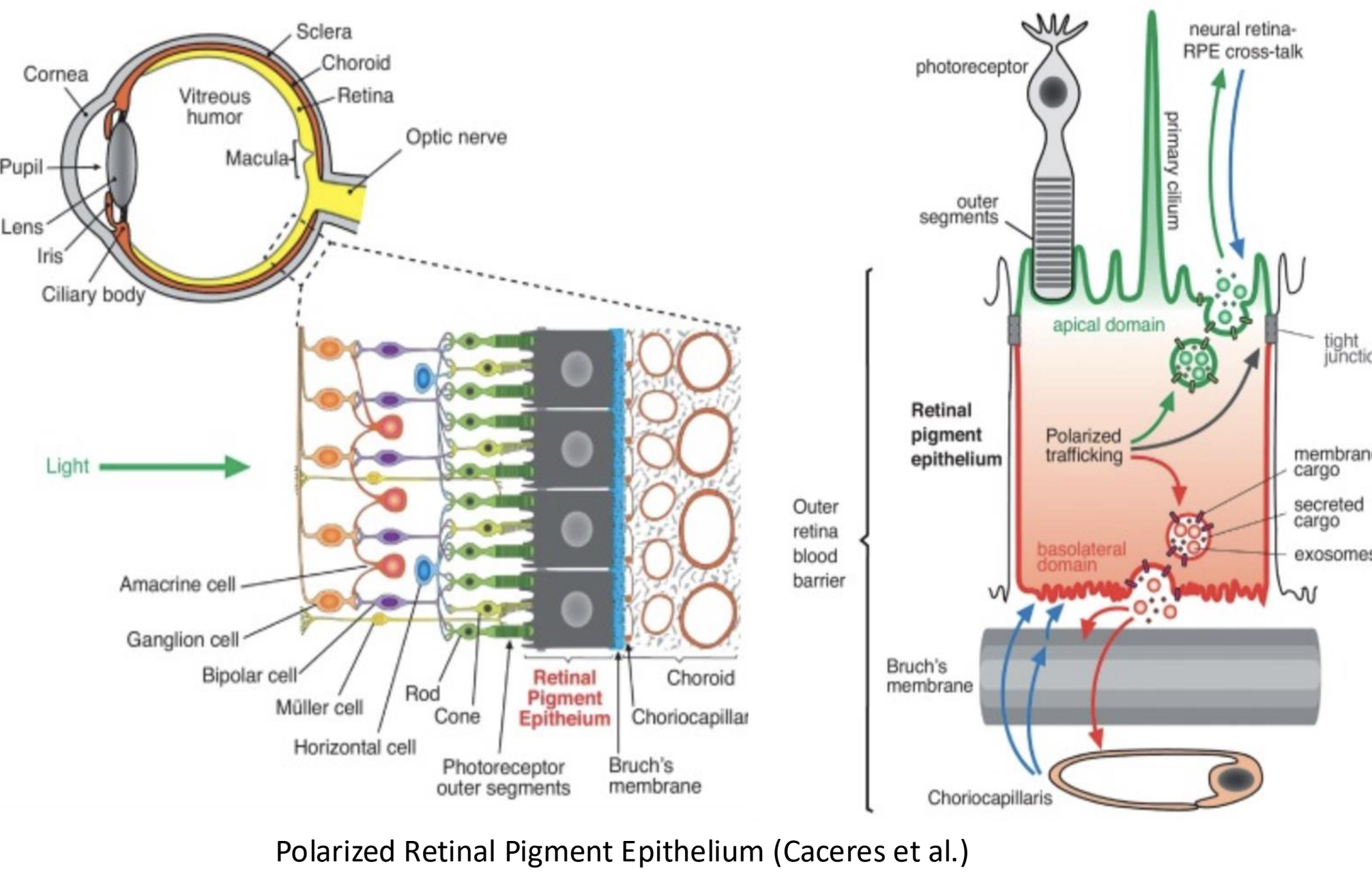


National Eye Institute

Tejjas Kaul<sup>1</sup> and Dr. Colby Lewallen<sup>2</sup>

1. University of North Carolina at Chapel Hill, Chapel Hill, NC; 2. National Eye Institute Ocular and Stem Cell Translational Research Section, Bethesda, MD

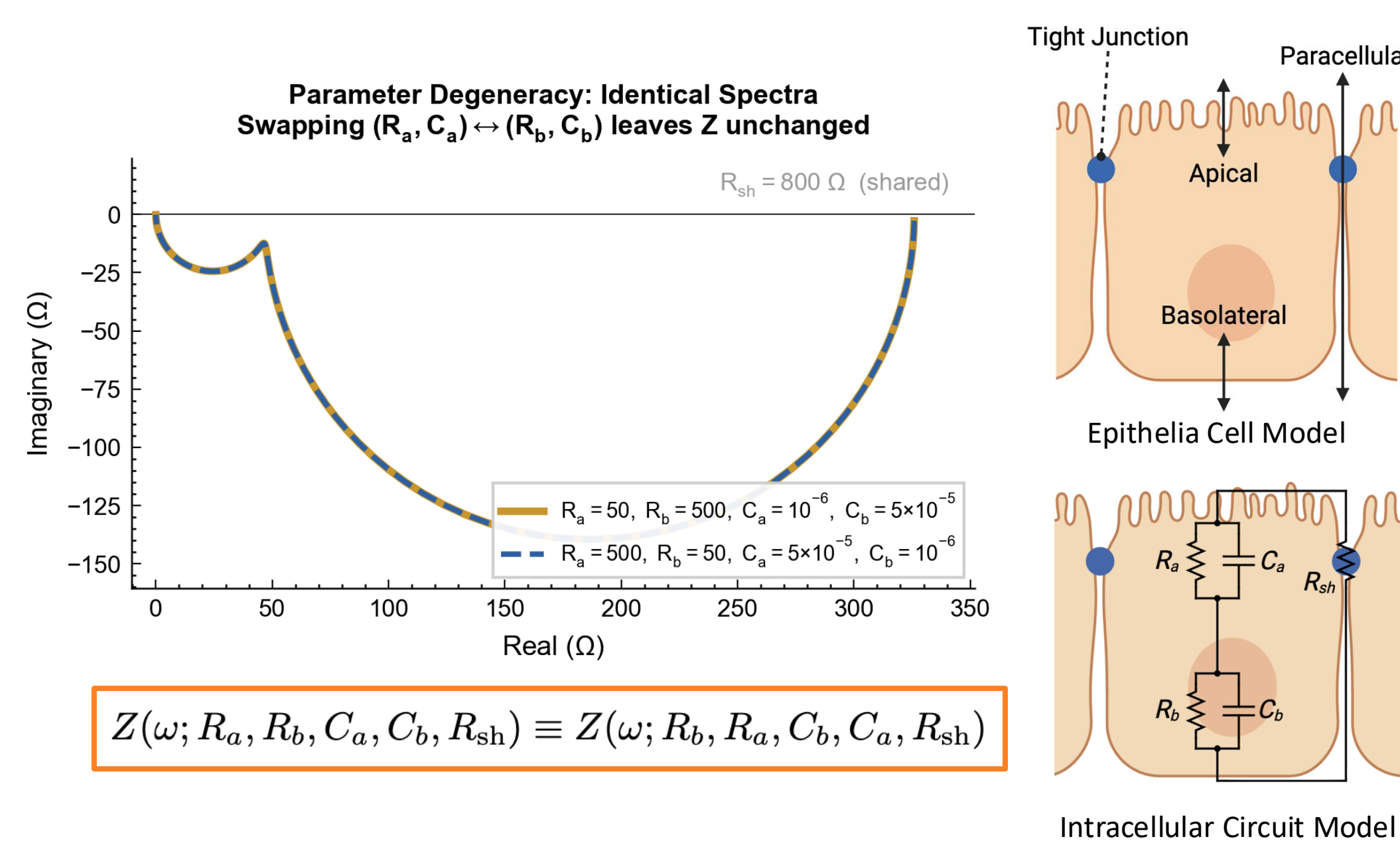
## Abstract



Epithelial membranes regulate cellular transport and serve as selective barriers against physical, chemical, and pathogenic damage. Hindrance of epithelial pathways is associated with conditions such as cystic fibrosis, diabetes, and age-related macular degeneration. Utilizing electrochemical impedance spectroscopy (EIS) to characterize electrical regions within epithelia is crucial for understanding membrane integrity and ion transport in response to drugs and disease. While conventional non-invasive methods yield electrophysiology measurements such as transepithelial resistance (TER) and capacitance (TEC), they lack multi-modal context and cannot reliably differentiate between apical, basolateral, and paracellular pathways.

We introduce a deep learning (DL) framework that augments the discovery of membrane circuit parameters, particularly for the retinal pigment epithelium (RPE), within an underdetermined equivalent circuit model. A physics-informed transformer architecture was trained on plausible synthetic impedance spectra generated with the equivalent circuit model and logarithmically scaled frequencies spanning 0.1 Hz to 1 MHz. Regularized to ensure biologically reasonable parameter predictions and consistency with known circuit theory, the model (5M parameters) inversely maps impedance to five circuit parameters: apical and basolateral resistances ( $R_a$ ,  $R_b$ ), apical and basolateral capacitances ( $C_a$ ,  $C_b$ ), and shunt resistance ( $R_{sh}$ ).

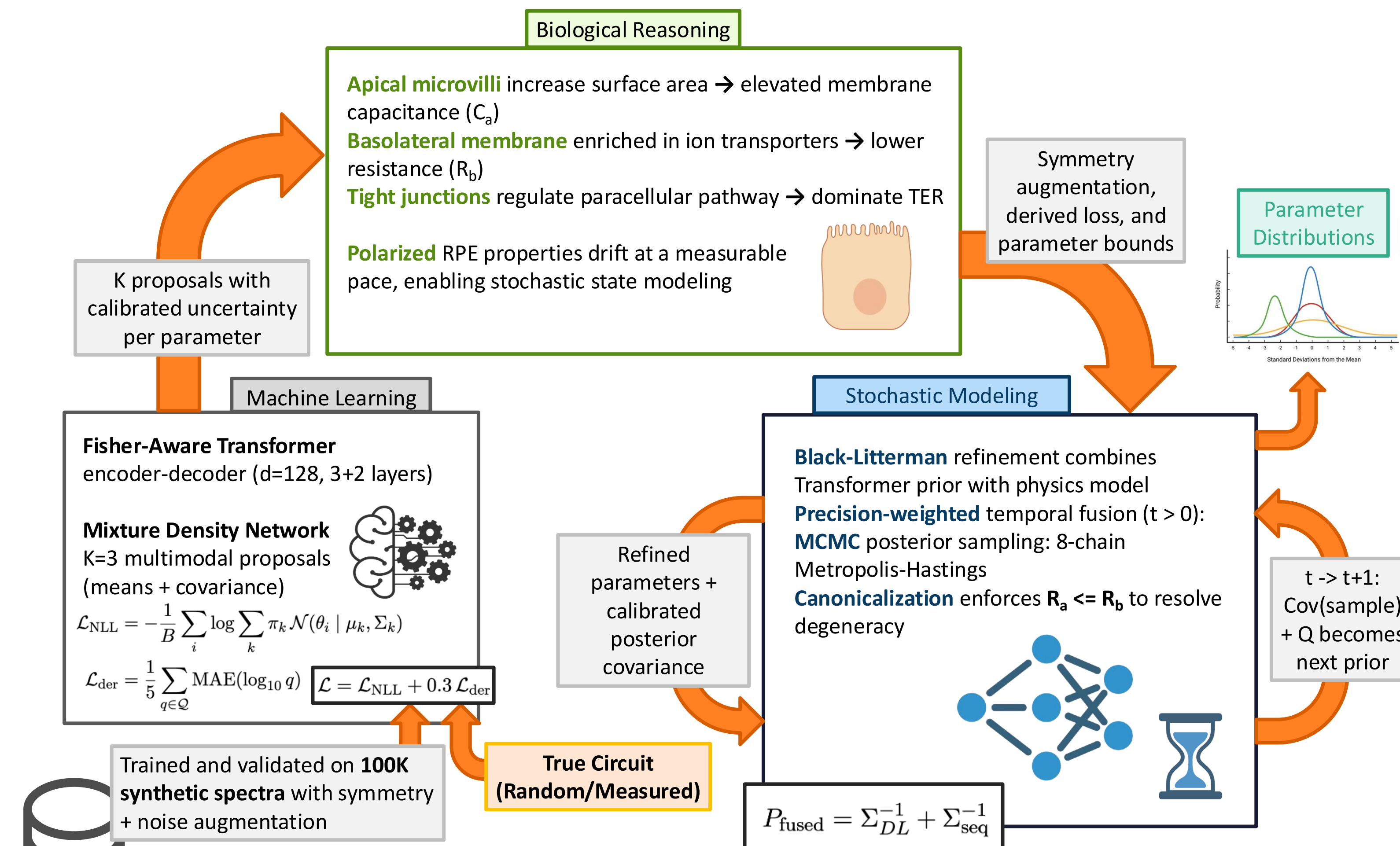
Preliminary testing on synthetic RPE circuits shows promising parameter recovery with heteroscedastic uncertainty quantification across simulated RPE perturbations while maintaining temporal consistency, a capability unavailable to traditional per-spectrum optimization approaches. Future work aims to incorporate evolutionary large-language model (LLM) cascades as dynamic rule proposers to discover constraint sets conditioned on impedance features, biological testing, and Nyquist morphologies; thus, tightening parameter-space variation and reducing parameter error.



## Methodology

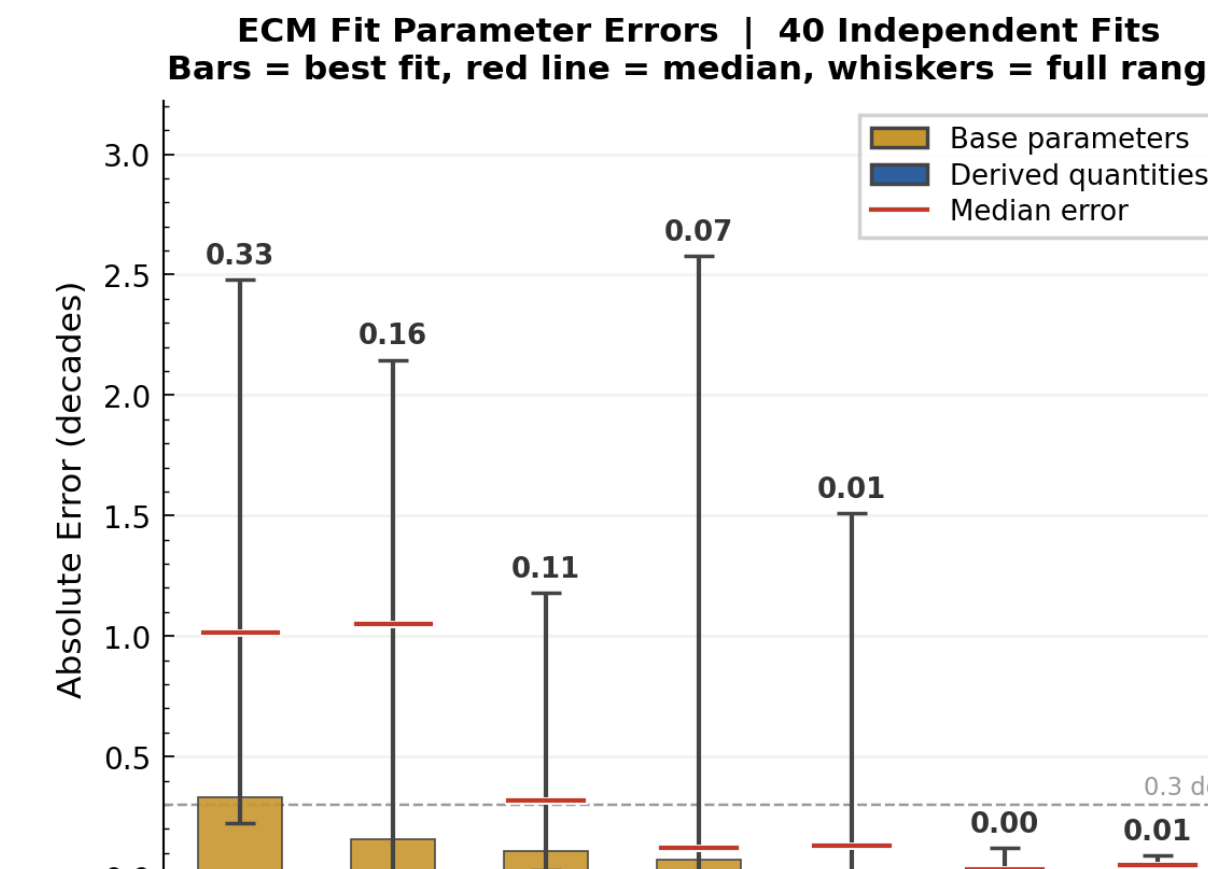
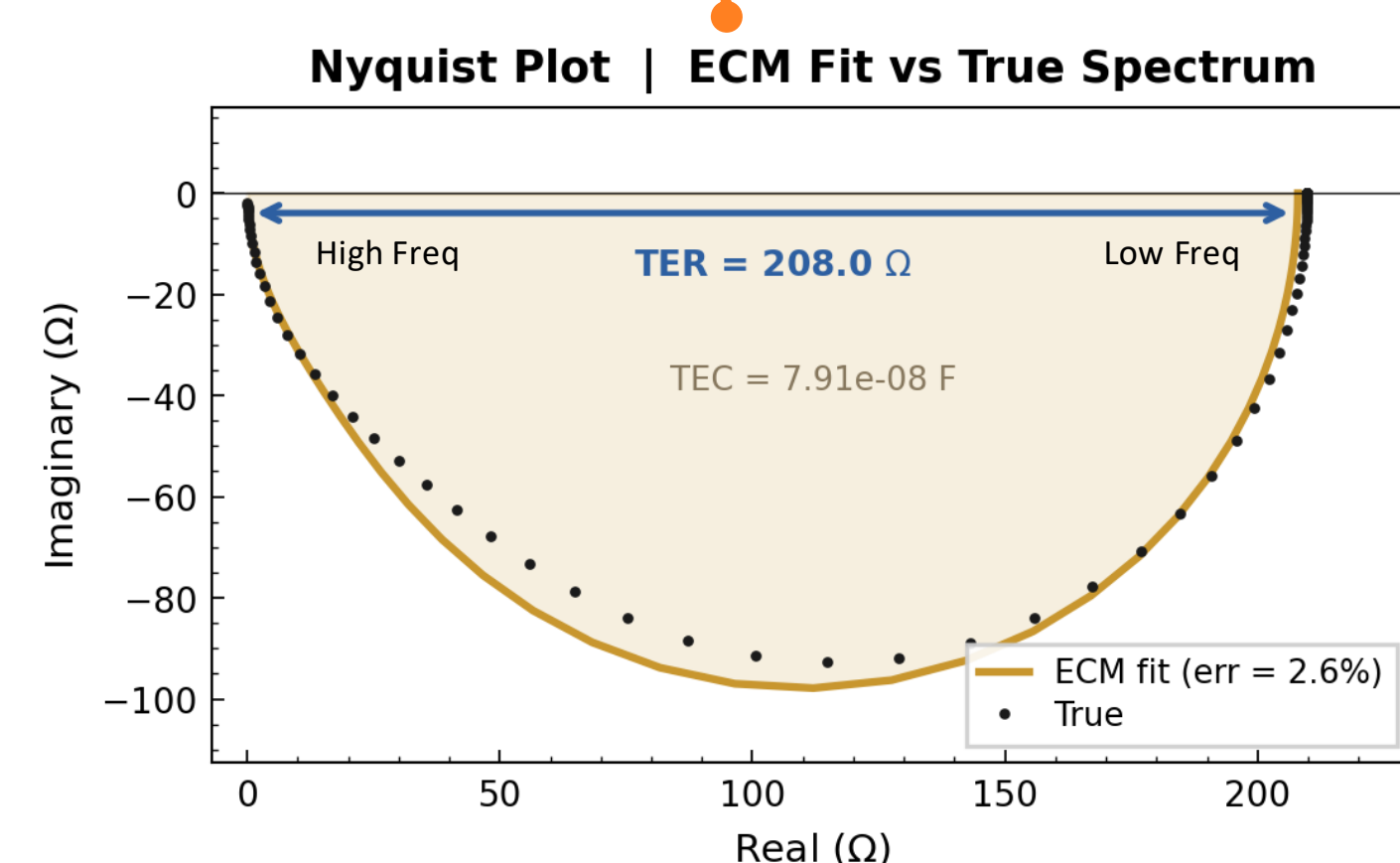
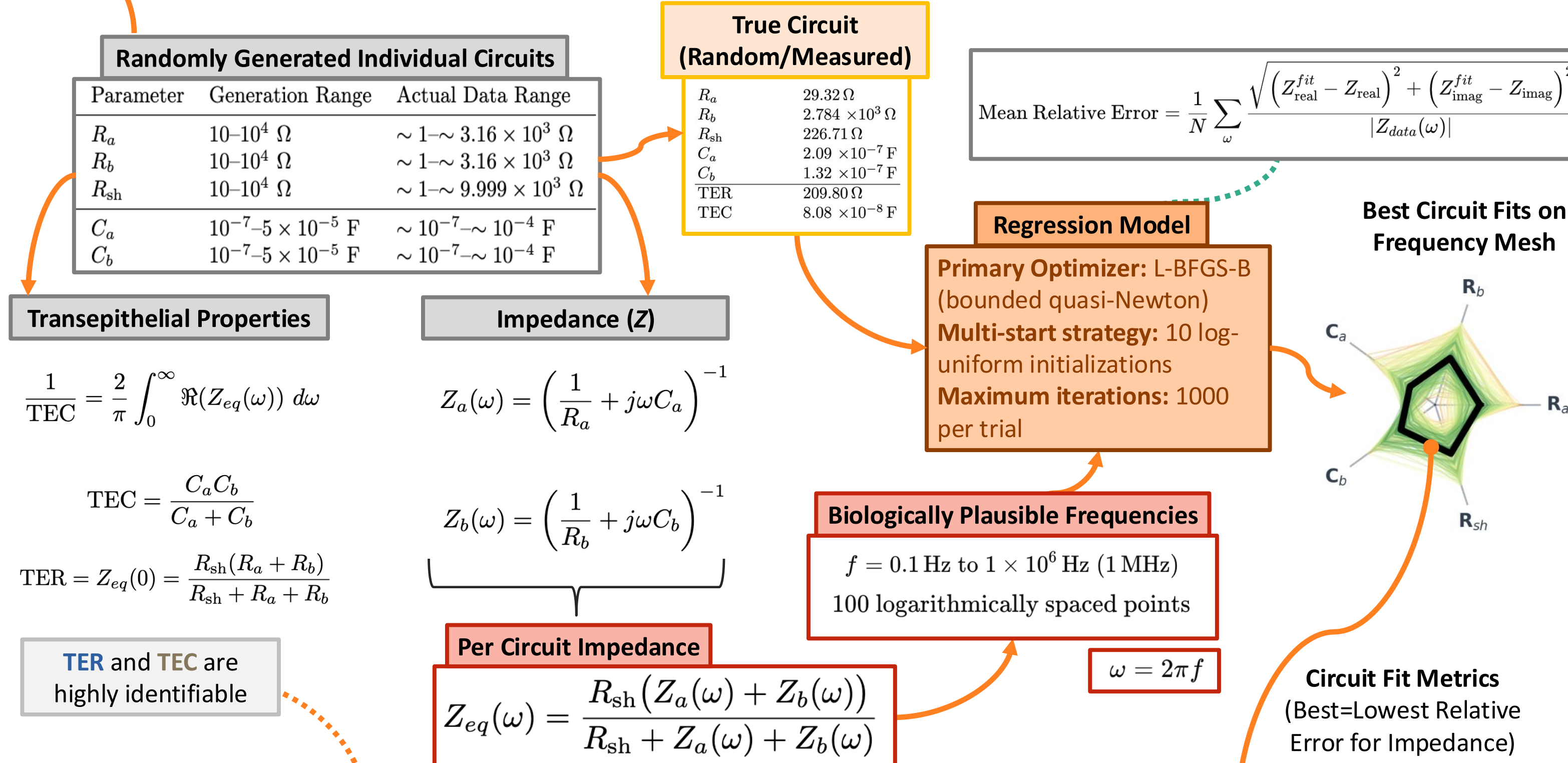
### Proposed Deep Learning Framework

Intertwining biological insights with machine learning and stochastic modeling yields better insight into RPE cells.

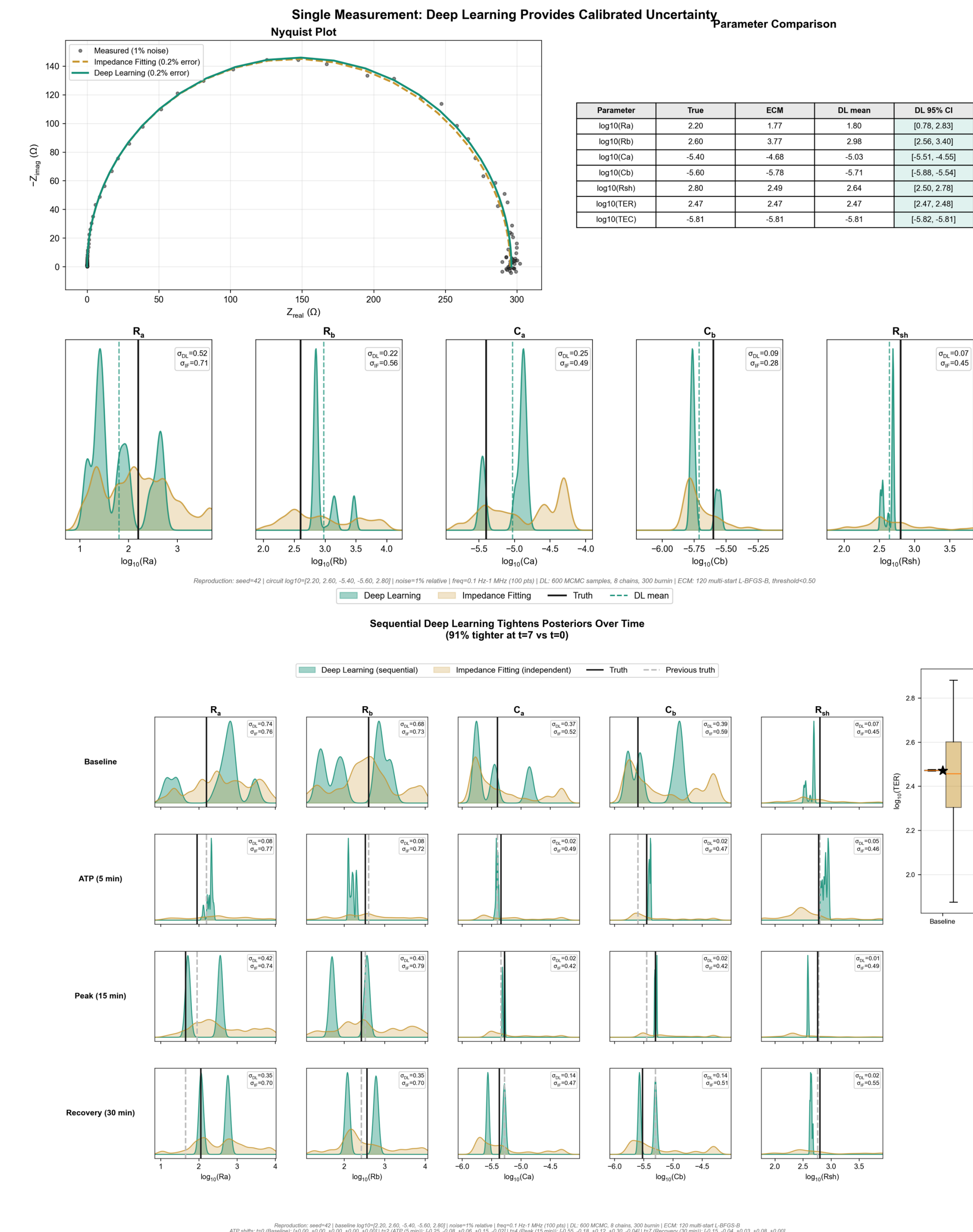


### Current Underdetermined Fitting Process

Fits are unpredictable since multiple parameter sets produce identical spectra; uncertainty is high and immeasurable.

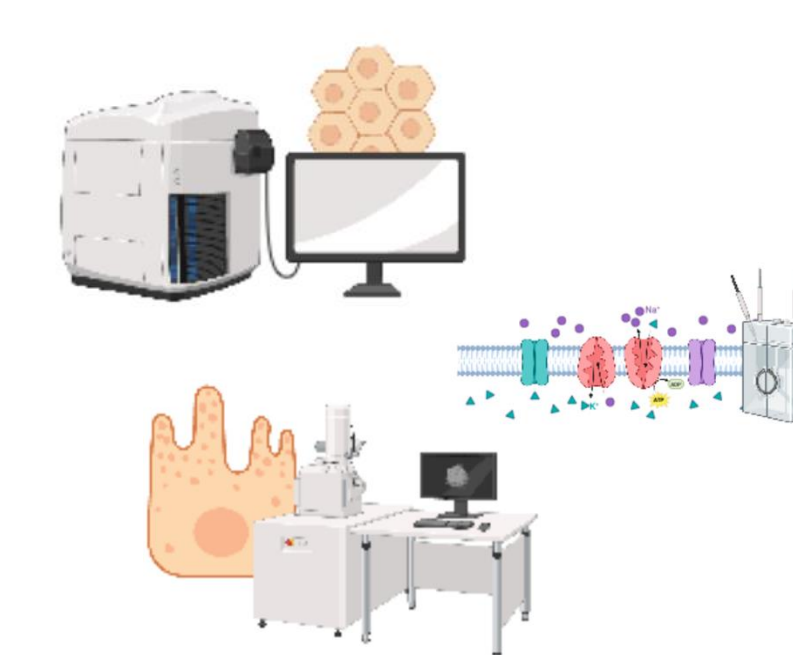


## Results

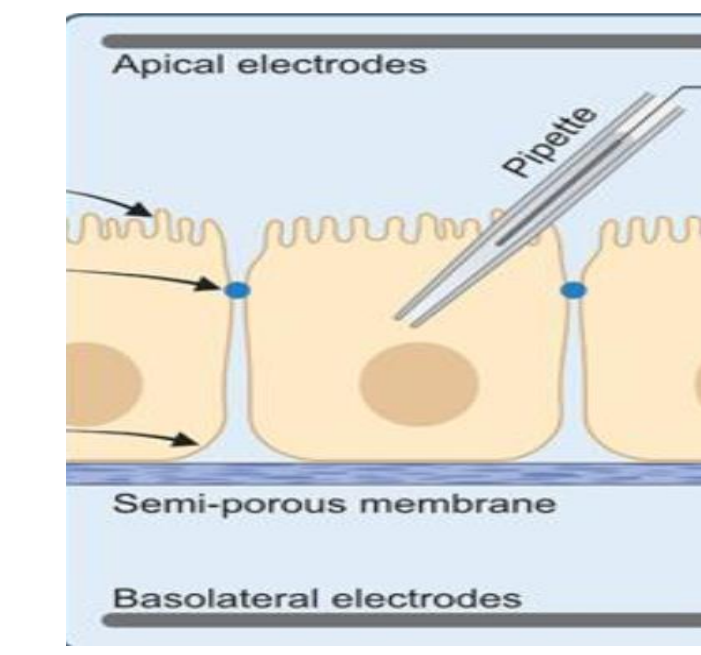


## Conclusion

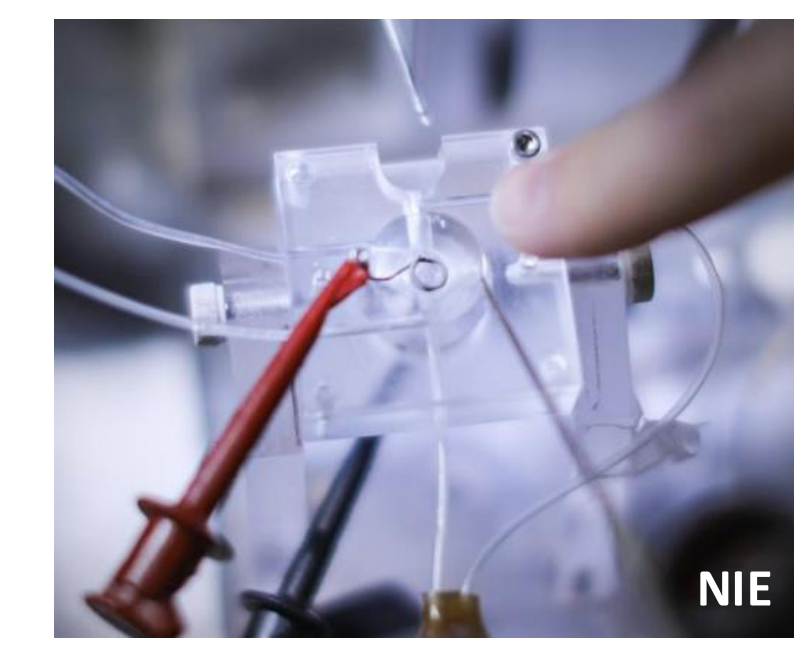
### Gold Standard



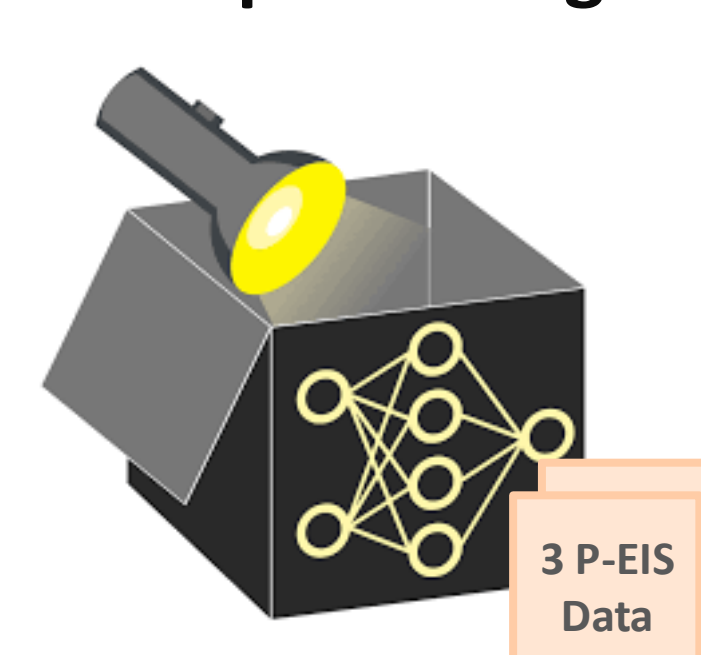
### Practical



### 3 P-EIS



### Deep Learning



High Throughput + Expensive

Low Throughput + Invasive

High Throughput + Non-Invasive

Interpretable + Refined

## Acknowledgement

We are grateful for the research group of Dr. Kapil Bharti at the Ocular and Stem Cell Translational Research Section, Ophthalmic Genetics and Visual Function Branch, National Eye Institute within the National Institutes of Health for culturing and providing mature iPSC-RPE cells and media. Figures created with BioRender. Interpretable ML concepts adapted from Bischl, B. et al., Interpretable Machine Learning, LMU Munich. Additional inspiration for the models was derived from \_\_\_\_.